

Privacy-Enhanced Federated Learning for Rare Genetic Disorder Classification with EHR

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Abstract—Utilizing machine learning and deep learning for disease classification has gained widespread popularity, yet it encounters significant challenges. Rare genetic disorders pose a data scarcity issue, as individual hospitals often lack sufficient data for accurate modelling. Collaborative data aggregation for such rare diseases raises data privacy concerns. Additionally, complex domains like genetic disorders require multifaceted data, including family history, medication history, and medical records. To address these challenges, this research explores federated learning on electronic health records (EHR) for the classification of rare genetic disorders. The study employs a dataset with a diverse range of patient information and leverages neural networks in a federated learning setup. The results demonstrate a 6% improvement in accuracy compared to individual hospital models, highlighting the effectiveness of this approach. Furthermore, the study provides insights into the trade-off between accuracy and privacy, emphasizing the importance of federated learning in safeguarding patient data while enhancing disease classification accuracy in the context of rare genetic disorders.

Keywords— *Classification, Electronic Health Records, Federated Learning, Genetic Disorder, Neural Networks, Privacy*

I. INTRODUCTION

In an age of advancing medical research and data-driven healthcare, the quest to understand and characterize rare genetic disorders is both vital and challenging. This research paper explores a privacy-enhanced approach – federated learning, shedding light on how it can revolutionize the study of these conditions using electronic health records.

A rare disease is a medical ailment that impacts a relatively small portion of a population. The specific criteria for defining something as "rare" may differ from one country to another, but it is typically determined by a low prevalence rate. As per the Orphan Drug Act [1], a rare disease is defined as a medical condition that affects fewer than 200,000 people in the United States.

Rare genetic disorders are medical conditions arising from anomalies or mutations in an individual's genetic material (DNA) and are distinguished by their infrequent occurrence within the population. Health analytics and prediction for rare genetic diseases indeed encounter formidable obstacles. Genetic diseases can be inherited from one's parents or can occur due to spontaneous mutations. The paucity of data, stemming from the small patient pool and fragmented datasets, greatly constrains the development of precise predictive models. Concerns regarding data privacy and ethical quandaries related to the use of genetic

information add significant complexity to the task. Additionally, the limited understanding of rare diseases, encompassing genetic variability and environmental factors, further hampers prediction efforts. Small sample sizes and constrained resources exacerbate these challenges, impeding progress in this crucial field.

To conduct effective analysis and prediction for rare genetic diseases, an extensive array of information is required. This encompasses not only genetic data but also comprehensive patient profiles, including family medical histories, health vitals, clinical observations, and lifestyle factors. The richness of this data enables researchers and healthcare professionals to develop more holistic and accurate predictive models. By combining genetic insights with a broad spectrum of health-related information, it becomes possible to unravel the intricate web of factors contributing to rare genetic diseases, potentially leading to earlier detection, tailored interventions, and improved outcomes for affected individuals. To advance in this field, collaborative efforts are essential, necessitating improved data sharing and stringent privacy protection.

This comprehensive data can be accessed through Electronic Health Records (EHR). An Electronic Health Record (EHR) is a digital representation of a patient's complete medical history, including their personal and treatment history, laboratory test results, medications, allergies, demographic information, and other essential health-related data[2]. EHRs are typically maintained and updated in a digital format by healthcare providers, such as hospitals and clinics, to facilitate the storage, retrieval, and distribution of patient details among healthcare professionals and organizations. The goal of EHRs is to improve the quality, productivity, and coordination of healthcare services while enhancing patient safety and accessibility to their medical information.

Electronic Health Records (EHRs) offer a powerful foundation for healthcare analytics and predictions related to rare genetic disorders. One significant challenge in this endeavour is the scarcity of data due to the rarity of these conditions. However, this obstacle can be effectively addressed through collaborative efforts and data aggregation. By combining data from various hospital EHRs and healthcare institutions, a more extensive and diverse dataset can be created, which enhances the robustness of predictive models. This pooling of data from multiple sources helps to overcome the limitations of individual datasets and provides a broader spectrum of patient information, ultimately

improving the accuracy and reliability of predictions for rare genetic disorders.

To mitigate data privacy concerns associated with sharing sensitive medical information, federated learning emerges as a promising solution. Federated learning (FL) is an advanced machine learning technique that allows models to be trained across decentralized datasets without the need to transfer the data itself. Instead of centralizing patient data, which could raise privacy and security risks, Federated learning allows the development of predictive models using data that is spread across various healthcare institutions or EHR systems. The advantage of federated learning is that it keeps patient data local, maintaining the privacy and confidentiality of individual records while still allowing for collective model improvement. This approach addresses data privacy issues by design, ensuring that sensitive patient information remains secure within each institution's database while assisting in the advancement of greater accuracy in predictive models for rare genetic disorders on a broader scale.

II. LITERATURE REVIEW

A. Electronic Health Records (EHRs)

1) *EHR Data Types*: EHRs encompass a wide array of data types and sources, including but not limited to : Subject of care identification, Demographics, Health history, Clinical summaries, Problem lists and diagnoses, Diagnostic values and interpretation, Care plans and decision support, Treatments, Consent records, Clinical documentation for chronic diseases, Encounters, Immunizations, Primary care and community care, Vital signs and alerts, Provider identification, Quality and safety information [3]. This comprehensive data compilation is invaluable for patient care, research, and healthcare management.

2) *Challenges and opportunities in utilizing EHR for research*:

a) *Data Privacy and Security*: Healthcare facilities face the daunting task of collecting, processing, and storing EHR data while safeguarding patient confidentiality. Regulations such as HIPAA in the United States and other country-specific privacy laws impose strict standards to ensure data security and privacy. In 2019, India's Union Cabinet approved a data protection bill that aims to require explicit consent for processing sensitive personal data and restrict unauthorized access by other organizations. Then came the Information Technology Act of 2000, under which digitally stored and transmitted data could be protected through Data Privacy Rules, primarily focusing on the protection of Sensitive Personal Information (SPI), encompassing a wide range of health-related data, such as psychological, physical, and mental health conditions, biometric information, patient history, medical records, and other sensitive information [4].

b) *Data Integrity Issues*: Senior employees, accustomed to traditional paper-based practices, frequently exhibit resistance to the successful adoption of EHR systems [5]. Inconsistent record-keeping practices, with some individuals choosing to document information digitally while others do not, can lead to data integrity and continuity issues within the system. This lack of uniformity hinders comprehensive healthcare management and decision-

making. To add on, missing data, poor record-keeping practices, and system losses can compromise the integrity of EHR data. Ensuring data accuracy and completeness is critical for reliable healthcare decision-making [6].

c) *Interconnectivity*: Existing practice systems often encounter challenges related to interconnectivity and standardization [6]. These issues arise from the diversity of EHR software implementations and inconsistencies in data formats, making seamless data exchange and integration problematic.

B. Rare Genetic Disease Prediction

1) *Previous Approaches* : In the context of predicting rare genetic diseases, diverse methodologies have been applied in the past, each with its specific strengths and drawbacks. One notable technique involves the use of Artificial Neural Networks (ANN) with Fog Computing, demonstrating an accuracy rate of 87% [7]. This approach highlights the significance of continuous knowledge enhancement and evaluation of the ANN model, with the potential for further accuracy improvement with increased knowledge.

Additionally, support vector machine (SVM) and K-Nearest Neighbours (KNN) models have been proposed, achieving remarkable results. The SVM model showcased training accuracy of 92.8% with a low misclassification rate of 7.2%, while the KNN model achieved similar training accuracy and miss classification rates [8]. During the prediction phase, the SVM model maintained an impressive accuracy of 92.5% with a 7.5% misclassification rate, and the KNN model exhibited 91.2% accuracy with an 8.8% misclassification rate [8]. These findings underscore the potential of machine learning in genetic disorder prediction.

Furthermore, recent advancements have introduced deep learning techniques. One study introduces AGDPM, a novel model, and utilizes the pre-trained AlexNet for predicting genetic disorders with genome data. AGDPM outperforms AlexNet with an 81.25% prediction accuracy, especially in single-gene, mitochondrial, and multifactorial inheritance disorders. The AGDPM model holds promise for enhancing genetic disease prediction in biomedical research [9].

To improve predictive accuracy, a study employed a chain classifier approach, utilizing multiple machine learning models equal to the number of classes. In this setup, each classifier utilizes the predictions generated by the preceding models to inform its own prediction. The study incorporated eight distinct machine learning models for performance assessment, including logistic regression, multi-layer perceptron, decision tree classifier, random forest classifier, KNN, extra tree classifier, extreme gradient boosting, and support vector classifier [10]. Fine-tuning of hyperparameters and data balancing techniques were implemented to optimize model performance and reduce the risk of overfitting.

2) *The importance of early detection* : In a broader context, the importance of early detection cannot be understated, particularly in genetic disorders like Alzheimer's. Detecting diseases and health conditions in their early stages allows for timely intervention and treatment, often leading to better outcomes and improved prognosis. Early detection not only enhances a patient's

chances of recovery but also helps in reducing the overall burden on the healthcare system and healthcare costs.

Despite extensive efforts to develop disease-modifying therapies for Alzheimer's disease (AD), the degenerative process remains challenging to halt [11]. AD is recognized as a chronic disease, with neuropathological changes occurring in the brain over a span of 10 to 20 years before clinical symptoms manifest [12]. This presses the need for early detection in genetic disorders as it could lead to better patient care. Early diagnoses also enable the initiation of therapies aimed at slowing disease progression and improving management [13].

Early diagnosis of individuals with a family history of genetic disorders has significant benefits to them and their families. It provides essential health information, aiding in tailored medical management and informed decision-making regarding family planning, prenatal care, and early intervention. For individuals facing untreatable gene disorders like Huntington's disease, knowing their carrier status can guide crucial decisions about childbearing, marriage, and financial security. Moreover, research shows that early detection can alleviate psychological stress associated with uncertainty and potentially enhance overall well-being. Notably, even in cases of positive test results, long-term adverse psychological effects appear to be limited [14].

C. Federated Learning

1) *Principles & Advantages* : Federated learning operates on several fundamental principles that enable distributed, privacy-preserving, and collaborative machine learning. First and foremost, it embraces decentralization, where multiple local devices or nodes each hold their own data and contribute to the model training process. This decentralized nature ensures that sensitive information remains on the device.

It works by enabling a central server to harness insights from diverse local models across various organizations, each contributing data from its specific population segment [15]. This diversity is magnified in healthcare when healthcare facilities span across different geographic regions. Hospitals situated in different geographic regions may gather diverse sets of data based on their location, and this geographic variability can also lead to patients from various regions displaying distinct characteristics, leading to varied data characteristics and sample sizes. This diversity underscores the value of federated learning, as models trained solely on local data within one medical organization may not deliver accurate predictions due to these differences. In such situations, a collaboration like such leads to the creation of a more robust and comprehensive shared model, ultimately enhancing prediction accuracy.

Another critical principle is the preservation of privacy. Federated learning employs techniques like differential privacy and encryption to protect individual data during the model training process. This ensures that personal information is never exposed, making it a crucial approach for a sensitive and regulated domain like healthcare.

Federated learning embodies these principles to enable efficient, privacy-conscious, and collaborative machine learning across a network of distributed devices or organizations.

2) *Applications in healthcare and genetics*: Federated learning, a groundbreaking approach in the realm of machine learning, has found compelling applications in healthcare despite its slow adoption rate. This innovative technique leverages the collaborative power of distributed datasets from multiple healthcare institutions and genetic research centres, paving the way for enhanced predictive models, privacy preservation, and the potential to revolutionize diagnosis and treatment in these critical fields.

In a particular study, an algorithm outperformed other models, achieving accuracies of 75.8% and 30.5% in binary and multi-class settings, respectively [16]. The study was able to identify factors that drop the model performance while federated averaging, such as lesser data from certain clients compared to the rest, as well as solutions to tackle the same.

Existing literature tackles the challenges posed by heterogeneous electronic healthcare record (EHR) systems and not independent and identically distributed (non-i.i.d) data in federated learning. In machine learning, a common assumption is that data is generated independently and identically (i.i.d.), implying that the process responsible for generating new samples has no recollection of previous ones. They introduce the UniHPF framework, a novel approach that unifies diverse EHRs within federated learning, marking the first integration of such heterogeneous EHRs in a single framework. The combination of UniHPF and federated learning yields an average performance boost of 3.4% compared to local learning methods. The authors assert its applicability in real-world FL scenarios [17].

An interesting study delves into the applicability of federated learning in the realm of genomic data, with the goal of encouraging its adoption in this domain. Although federated learning has gained recognition in clinical data applications due to its data privacy advantages, its effectiveness in genomics is a relatively unexplored area. The research investigates two scenarios: phenotype prediction using the UK Biobank data and ancestry prediction based on the 1000 Genomes Project data. The findings demonstrate that federated models, operating on independently partitioned data nodes, achieve performance levels akin to centralized models, even when faced with considerable inter-node disparities [18]. The study also illustrates how federated prediction models can seamlessly integrate with data processing steps like federated PCA, thus advocating for the widespread adoption of federated models in healthcare.

D. Privacy and Security in Healthcare

1) *Privacy concerns with patient data*: Privacy in patient data is crucial to protect individuals' sensitive information and maintain their trust in healthcare systems while complying with legal regulations like IT Act, GDPR and more personal data regulatory laws [4]. Privacy and security issues related to electronic health information can be broadly categorized into two areas [19]. Firstly, there are concerns about the improper disclosure of information within individual organizations, encompassing both internal and external threats. Internal threats involve authorized system users abusing their access privileges for unauthorized purposes, such as viewing the records of

acquaintances or leaking information. External threats consist of unauthorized outsiders attempting to access or manipulate data.

In the past, healthcare institutions primarily concentrated on mitigating internal risks to secure physical health records, but they have had limited exposure to safeguarding digital health data from external technical threats. This is primarily due to the less prevalent connectivity to public networks until recent times. Secondly, concerns of a systemic nature revolve around the potential misuse of patient-specific health information, which could hinder an individual's employment opportunities, access to insurance, or even lead to personal embarrassment [19]. To better grasp the origins of these concerns, it's important to examine how health information flows throughout the healthcare system.

Existing privacy techniques: Implementing privacy techniques in healthcare is paramount as it ensures the protection of sensitive patient information, prevents unauthorized access, and builds trust between healthcare providers and patients. This entails the implementation of robust security measures, encryption protocols, and access controls to ensure the protection of electronic health records and the preservation of patient confidentiality. These steps are vital in the current age of healthcare digitization and transformation.

One study describes privacy preserving methods while generating models with health data in great depth [20]. Firstly, they describe the need for privacy-preserving model generation solutions to prioritize the prevention of private information leakage during the training process. This can be achieved by minimizing sensitive information from the training data.

Regarding data-centric privacy measures, existing approaches focus on several directions: the application of traditional anonymization methods (Ex. k -anonymity, l -diversity, and t -closeness) to remove identifier information from the training data, representing the raw dataset through surrogate datasets or implementing differential privacy mechanisms thus mitigating private information leakage.

Additionally, the paper provides detailed insights into several models designed to implement these privacy techniques effectively, addressing both data and computation aspects to ensure the protection of sensitive information.

III. METHODOLOGY

Our methodology leverages federated learning to collaboratively train a machine learning model across distributed healthcare institutions while safeguarding data privacy. By aggregating insights from electronic health records, we empower accurate classification of rare genetic disorders while safeguarding sensitive patient information.

A. Dataset

The dataset selected for this study is the "Of Genomes And Genetics" dataset [21] obtained from Kaggle, which encompasses a diverse collection of genetic disorder-related data, providing a valuable resource for our research. The dataset comprised information from 22,083 patients, encompassing 45 distinct fields of data.

The target variable for this study was the subclassification of genetic disorders, which encompassed nine subclasses, consisting of six rare genetic disorders and

three non-rare genetic disorders. Among the rare genetic disorders were 'Mitochondrial myopathy,' 'Tay-Sachs,' 'Cystic fibrosis,' 'Leigh syndrome,' 'Leber's hereditary optic neuropathy,' and 'Hemochromatosis,' while the non-rare genetic disorders included 'Alzheimer's,' 'Cancer,' and 'Diabetes.'

The set of independent variables comprised 43 distinct fields encompassing diverse elements extracted from the Electronic Health Records (EHR). These fields encompassed aspects such as medical history, medication history, genetics, family history, and personal history. Subsequently, employing a series of Exploratory Data Analysis (EDA) techniques, elaborated upon later in this document, we identified and extracted the most valuable features.

B. Exploratory Data Analysis (EDA) and Preprocessing

The data underwent thorough EDA and preprocessing steps. Initially, irrelevant columns were removed, missing data was handled by imputing values based on data grouping, and essential features were selected through correlation analysis. Outliers were detected and dealt with using the Interquartile Range (IQR) rule. Feature encoding involved one-hot encoding for 'Disorder Subclass,' label encoding for categorical data, and averaging for age, symptom, and test features. Data normalization was performed with Min-Max scaling for uniform feature scaling. To address class imbalance, SMOTE was used. Finally, for model training and evaluation the dataset was split into training, testing, and validation sets.

C. Model Selection

Following the completion of data preprocessing, our task was to select a machine learning model that optimizes accuracy while minimizing recall, all while taking into account computational efficiency and memory utilization.

After conducting extensive research by reviewing multiple articles and research papers [22][23][24][25], we meticulously identified a pool of five machine learning algorithms best suited for the efficient and accurate classification of genetic disorder subclasses. In Table I., we present the machine learning algorithms we trained, along with their respective accuracy and F1 score, enabling us to make informed decisions when selecting the most proficient model for our study. We opted for the Sequential Neural Network based on its accuracy performance.

TABLE I. MODEL EVALUATION

Sr. No.	Algorithm	Accuracy	F1 Score
1	SequentialNeuralNetwork	0.76	0.76
2	XGBClassifier	0.74	0.74
3	BaggingClassifier	0.67	0.66
4	RandomForestClassifier	0.67	0.66
5	LogisticRegression	0.50	0.50

D. Hyper Parameter Tuning

Our approach for systematically exploring the hyperparameter space and determining the optimal

configuration for our sequential neural network relies on the GridSearchCV algorithm. GridSearchCV is a widely adopted technique that conducts a thorough examination of a predefined grid of hyperparameters. This grid encompasses diverse sets of hyperparameter combinations, and for each combination, the algorithm conducts model training and evaluation through cross-validation.

The hyperparameters that are pertinent to our sequential neural network encompass the following:

Batch size

Number of epochs

Using a range of values, we set the hyperparameters and the optimization evaluation metric, such as the accuracy and F1-score. Subsequently, GridSearchCV methodically explores all conceivable hyperparameter combinations, engaging in model training and performance assessment utilizing k-fold cross-validation.

The optimal hyperparameters for our sequential neural network were found to be:

Batch size: 25

Number of epochs: 100

We contrast the initial, untuned model's performance with that of the model running with the ideal hyperparameters. The evaluation measures that are employed are F1-score, recall, accuracy, and precision. The model's performance significantly improves, which emphasizes the significance of hyperparameter adjustment.

E. Neural Network Setup

The neural network, referred to as “model3,” serves as the foundation for our experiments and findings. We will discuss the architectural choices, layer configurations, and key components of this neural network setup.

Our neural network, model3, is constructed using the Keras framework. Being a sequential model, the layers are added one after the other linearly. This architecture is well-suited for a wide range of tasks, including classification, which is the focus of our research.

Model3 comprises several layers, as shown in fig 1., each designed to capture and transform the input data. The layer-by-layer configuration is as follows:

Input Layer: The input layer consists of 100 neurons, which receive data with 26 features. We apply the ReLU (Rectified Linear Unit) activation function, a commonly used choice for introducing non-linearity.

Hidden Layers: Model3 includes three hidden layers, each with a different number of neurons – 48, 16, and 8, respectively. The model can identify complex patterns and representations from the input data with the aid of these hidden layers. These layers use the ReLU activation function, just like the input layer.

Output Layer: The output layer contains nine neurons, corresponding to the nine classes in our classification problem. The activation function used here is “softmax,” which is ideal for multi-class classification tasks. It transforms the network's raw outputs into probability distributions over the classes, allowing us to make class predictions.

Before training, it's essential to configure the model's learning process. In our case, we compile the model using the following settings:

Loss Function: The loss function that we select is “categorical_crossentropy.” This is suitable for multi-class classification problems, where the goal is to reduce the difference between the actual class labels and the predicted class probabilities.

Optimizer: We opt for the “adam” optimizer, a popular choice for gradient-based optimization. Adam combines the advantages of both momentum and RMSprop, offering efficient convergence during training.

Metrics: We make use of a variety of metrics, such as accuracy, precision, and recall, to evaluate the model's performance during training and evaluation. These metrics offer information about various facets of the model's predictive abilities.

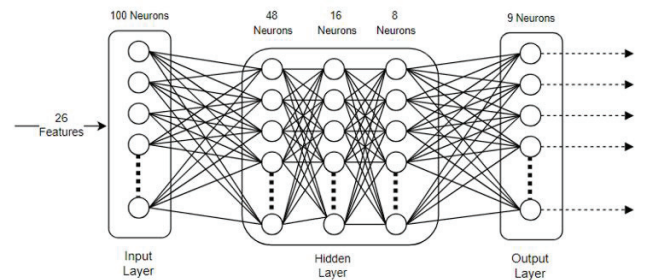


Fig. 1. Neural Network Architecture

F. Federated Learning Methodology

Now, federated learning is employed with a neural network to address two critical issues:

Enhancing accuracy in rare genetic disorder classification by aggregating data from diverse sources

Ensuring data privacy by allowing hospitals to collaborate without sharing sensitive patient information.

1) *Architecture:* The research utilizes a federated learning architecture where the dataset is partitioned among three client nodes, representing distinct hospitals.

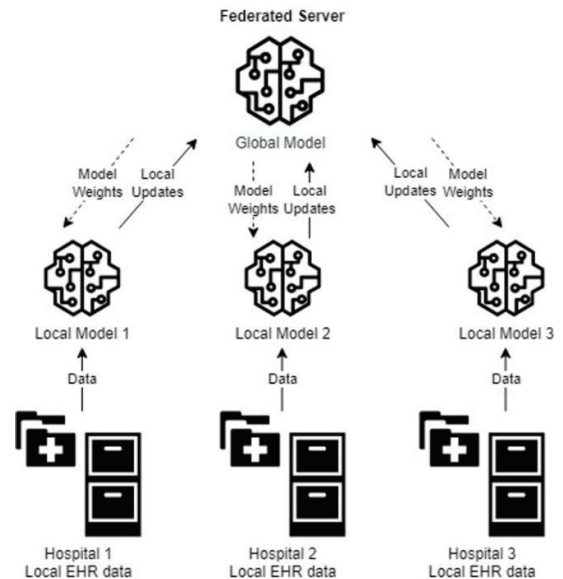


Fig. 2. Federated Learning Architecture

Each client trains an individual neural network on their local server, with local updates periodically aggregated at the central server, forming a global model. The updated model weights are then sent back to the clients by the global model, iteratively repeating this process for the selected number of rounds. This federated approach ensures collaborative model improvement while preserving data privacy. Fig 2. Shows the federated learning architecture.

2) *Flower*: For federated learning implementation, we have chosen the Flower framework. In the realm of Federated Learning, Flower stands out as an open-source framework, ideal for swift deployment and experimentation [26]. Flower provides a seamless infrastructure for federated learning by facilitating communication and coordination among the client nodes and the central server. It streamlines the process of federated model training, allowing us to efficiently orchestrate model updates, aggregation, and parameter synchronization across the distributed network of hospitals while ensuring the security and privacy of sensitive patient data. Flower's adaptability and robustness make it a suitable choice for our federated learning approach in this research.

3) *Federated Learning Process*: Fig. 3 explains the process of federated learning with a neural network:

a) **Initialization**: Each participating hospital (client) initializes its own neural network model with random weights and a central server initializes a global model with a common initial weight vector.

b) **Local Training**: On each client, the local neural network model is trained using its respective local dataset. Training is performed for a fixed number of local epochs using standard optimization techniques (e.g., gradient descent) to update the model's weights.

c) **Local Model Update**: Following local training, the client calculates gradient updates using its local dataset, without disclosing the actual data. Subsequently, these gradient updates are transmitted to the central server.

d) **Model Aggregation (FedAvg)**: The central server collects gradient updates from all clients. It aggregates these updates to calculate a new global model weight vector. The aggregation can be performed using techniques like Federated Averaging (FedAvg), where the global model's weights are updated as an average of the local updates from participating clients.

e) **Global Model Distribution**: The central server disseminates the revised global model weights to all engaged clients.

f) **Repeat Steps c-e**: Steps c to e are repeated for a predefined number of communication rounds or until convergence criteria are met. The global model gradually improves as clients collaborate to contribute knowledge while keeping their data locally.

g) **Evaluation**: Periodically or at the end of training rounds, the global model's performance is evaluated on a separate validation dataset or through other evaluation metrics.

h) **Termination and Deployment**: The federated learning process can be terminated when the desired model

performance is achieved or after a predetermined number of rounds. The final global model can be deployed for making predictions or further analysis.

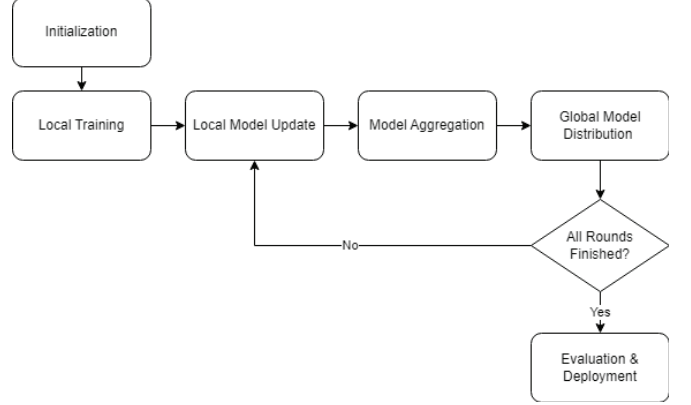


Fig. 3. Federated Learning Process

IV. RESULT AND DISCUSSION

In this section, we delve into the outcomes of three distinct approaches employed in our research, each addressing the pivotal challenge of enhancing model accuracy for rare genetic disorder classification while safeguarding patient data privacy.

A. Scenarios

1) *Scenario 1: Centralized Server* – In the first scenario, a centralized server was employed, where all the medical data from multiple hospitals was combined to train a single sequential neural network (Fig. 4).

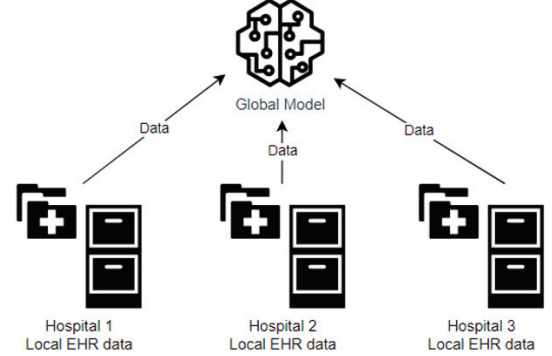


Fig. 4. Scenario 1

2) *Scenario 2: Individual Local Servers* – In the second scenario, individual hospitals trained their own neural networks independently using their local datasets (Fig. 5).

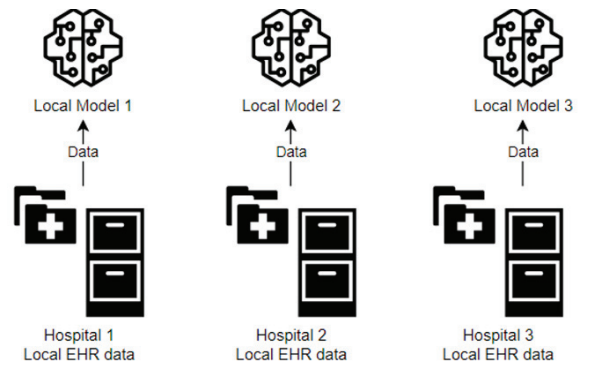


Fig. 5. Scenario 2

3) *Scenario 3: Federated Learning* – The third scenario adopted federated learning, leveraging the Flower framework. Here, participating hospitals maintained data privacy by keeping their patient records on-site and only sharing model updates (Fig. 2).

B. Results:

1) *Scenario 1:* In the first scenario, a centralized server was utilized to consolidate medical data from multiple hospitals, enabling the training of a unified sequential neural network. This approach yielded an accuracy rate of 76% in our rare genetic disorder classification task. However, it was marred by substantial privacy concerns related to Electronic Health Record (EHR) data. The necessity to amalgamate sensitive patient information from various sources raised significant ethical and legal concerns, rendering this approach unviable for real-world healthcare applications. Protecting patient privacy remains a paramount concern in medical data analysis, demanding alternative strategies that can mitigate these privacy risks.

2) *Scenario 2:* In the second scenario, we opted for a decentralized approach, where each hospital independently trained its own neural network using its local dataset. While this approach helped alleviate privacy concerns by keeping patient data within the confines of each hospital, it also unveiled some intrinsic challenges. Model accuracies, as indicated in Table II. below, varied significantly among hospitals due to differences in dataset sizes and heterogeneity in the patient population. This decentralized strategy presented limitations in achieving the desired level of model accuracy and consistency across all participating hospitals, highlighting the need for collaborative methodologies that can effectively harness the collective knowledge from multiple sources while respecting data privacy.

TABLE II. CLIENT EVALUATION

Algorithm	Accuracy
Client 1	0.51
Client 2	0.49
Client 3	0.56

3) *Scenario 3:* In our exploration of a balanced approach between data privacy and model performance, we engaged in the third scenario: the utilization of federated learning with the Flower framework. This approach effectively addressed the paramount concern of healthcare data privacy. By enabling hospitals to maintain control over their patient records while jointly training a global neural network model, we achieved a reasonable increase of 6% in the aggregated accuracy during rare genetic disorder classification. While this accuracy may fall slightly short of the centralized approach, it signifies a substantial step towards reconciling the trade-off between model performance and data confidentiality in healthcare analytics. Federated learning, as demonstrated, offers a practical and responsible solution for collaborative medical research and

diagnostics that respects the ethical imperative of patient data privacy.

V. CONCLUSION

To summarize, our paper delves into the intricate interplay between data privacy and model accuracy within the realm of healthcare analytics. This research is underscored by the pressing challenge of addressing rare genetic diseases, which are characterized by limited available data.

Our exploration of three distinct scenarios illuminates the potential and pitfalls of different approaches. The centralized approach, though achieving a commendable accuracy of 76%, is hampered by severe privacy concerns associated with the amalgamation of medical EHR data from diverse sources. On the other hand, the decentralized model training in Scenario 2, while preserving data privacy, suffers from disparate accuracies across hospitals, underscoring the limitations of an isolated approach.

In Scenario 3, we adopted federated learning with Flower, achieving a reasonable accuracy rate of 64%. This approach strikes a promising balance, respecting patient data privacy while still obtaining an acceptable level of model performance. It highlights the potential for responsible, collaborative rare genetic disease classification within the healthcare landscape, where both data security and diagnostic accuracy are paramount.

Furthermore, our research underscores the critical issue of data scarcity in the context of rare genetic diseases. Electronic Health Records (EHRs) emerge as a crucial resource, offering comprehensive patient-specific information that aids in classification and diagnosis. By pooling knowledge and leveraging EHRs across healthcare institutions, we address the inherent rarity of these conditions.

In conclusion, our study not only navigates the delicate equilibrium between data privacy and model performance but also underscores the urgency of combining knowledge to tackle data limitations in rare genetic disease research. EHRs, as valuable data sources, have the potential to bridge this gap and enable effective healthcare analytics. This collaborative, privacy-conscious approach holds significant promise in advancing our understanding of rare genetic diseases and improving patient care outcomes. As we move forward, it is vital to continue exploring innovative methodologies that balance the dual imperatives of data privacy and diagnostic accuracy in the pursuit of medical knowledge and better patient care.

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